

PROFESSIONAL SUMMARY

Staff AI Engineer at **EchoStar** building the AI pipelines behind consumer-grade telecom – retrieval, agent orchestration, and evaluation running against a network millions of customers depend on. **Ex-Apple** and **ex-Walmart** background deploying production-grade LLM and AI systems – **tripled backend throughput** on Java and Spring Boot microservices at Walmart and kept **macOS storage releases stable across 100M+ devices** at Apple. Replaced a vendor agent with one that has never written a wrong record across **10,000 support cases/month**, cutting cost per case from \$1.00 to ~\$0.02 (**saving ~\$120K/yr**). Ran on-prem open-weight models on **vLLM** behind carrier-scale platforms answering questions in **under 200ms** across billions of records.

SKILLS

Languages	Python, TypeScript, JavaScript, Java, Go, Rust, C/C++, SQL, Bash, HTML/CSS
Backend & APIs	Java/Spring Boot, Spring MVC, Hibernate/JPA, FastAPI, Flask, Django, Node.js/Express; REST, GraphQL (nested queries, mutations, resolvers), gRPC, WebSockets; microservices, serverless (AWS Lambda); distributed systems, system design, concurrency, caching, SOLID, design patterns, DDD
Frontend & Full-Stack	React, Next.js (App Router, SSR/ISR), TypeScript, Tailwind CSS, Redux/Zustand, Angular, responsive/accessible UI, Supabase, Stripe payments and webhooks
AI / ML Engineering	vLLM, PyTorch, scikit-learn, LangChain, LangGraph, agentic workflows, agent loops and harnesses (Claude Code, Hermes, OpenClaw), context engineering, LLM fine-tuning (LoRA/QLoRA), MLOps and ML pipelines, RAG, embeddings, Vector DBs (pgvector, FAISS), Model Context Protocol (MCP), Pydantic, MLflow, LangSmith
Data & Streaming	Kafka (partitions, consumer groups, replay), RabbitMQ; event-driven and pub/sub streaming, schema validation (Avro, JSON Schema), real-time state sync; ClickHouse, Elasticsearch, Spark, Airflow, Postgres, MySQL, Oracle, Redis, MongoDB, DynamoDB, Snowflake, BigQuery; ETL pipelines, SQL query tuning, schema design
Cloud & DevOps	AWS (EC2, ECS, Lambda, S3, RDS, Bedrock, SageMaker), GCP, Azure; Kubernetes, Docker, Terraform, Ansible, Helm; CI/CD (GitHub Actions, Jenkins), Git, Linux, Agile/Scrum, code review
Testing & Reliability	pytest, JUnit, Playwright, Jest/Vitest; TypeScript and Node.js test frameworks and harnesses, GraphQL and REST contract validation in CI/CD; TDD, unit/integration/E2E testing, coverage gates, load testing; Prometheus, Grafana, OpenTelemetry, Datadog, SLOs, incident response
Security	OWASP Top 10, RBAC, IAM, OAuth 2.0, SSO/SAML, secrets management, least-privilege tool access, prompt-injection defense, tool-abuse guardrails

EDUCATION

- University of California, Berkeley** | *Bachelor of Arts, Computer Science; GPA 3.7*

Berkeley, CA | Aug 2019 – Aug 2023

PROFESSIONAL EXPERIENCE

- EchoStar** | *Staff AI Engineer* SF Bay Area | Sep 2026 – Present
 - Builds the AI Pipelines Behind a Consumer Telecom Network** Staff AI Engineer building the **retrieval, agent orchestration**, and **evaluation** systems behind EchoStar's consumer-grade network, carrying the production-agent and RAG patterns proven at Polaris and in independent consulting to a network **millions of customers** depend on.
- Independent Consulting (W3 Sourcing, ERIA, LUPFR)** | *Technical Partner & Software Engineer* San Francisco, CA | Mar 2026 – Present
 - Saved ~\$120K a Year on Support for a Major Water Filtration Company** Support was buried in cases and Salesforce Agentforce charged **\$1.00** to handle each one. Built a triage agent where the model suggests and plain Python decides what gets saved, so it has never written a wrong record across 10,000 cases/month. Cost per case fell to **\$0.02**, saving **~\$120K/yr** and returning **40 hours/week**, alongside a FastAPI, Next.js, and FAISS knowledge base the same team answers from.
 - Replaced Manual Revenue Work at a \$10M+/yr Business (ERIA)** An events company serving Anthropic, OpenAI, and J&J was running ticketing and revenue by hand. Lead engineer on the **React, Next.js**, and **Supabase** store replacing it, handling payments, refunds, and duplicate charges safely, plus an admin agent. Shipped behind **2,895 tests**. (www.eria.co)
 - Built an Agentic ATS That Replaced Manual Sourcing (W3 Sourcing)** Recruiters were sourcing, screening, and ranking by hand. Built the **TypeScript** & React ATS that scrapes and ingests candidates, then ranks and screens **19K+ profiles** against every open role, with a chat agent whose server checks each quoted claim against the real profile. Replaced **\$20K/yr** of vendor tooling and now feeds a sourcing pipeline the firm projects at **\$500K–\$1M/yr** in placements. (www.w3sourcing.com)
 - Handed a Non-Technical Team the Keys (LUPFR)** This team needed an engineer every time they published an event. Made one link do the whole job and handed over a **React & Node.js** portal they run themselves, with analytics and CSV export. (lupfr.com)
- Polaris Wireless** | *Platform & AI Software Engineer* Santa Clara, CA | Sep 2024 – Jul 2026
 - Cut Pre-Release Verification from Days to Minutes** Built an agent in **Node.js & Python** booking/freeing lab machines on request plus a pipeline generating test cases, taking coverage from 40 to **300+ scenarios** and pre-release verification from days to minutes.
 - Gave Analysts Direct Access to Carrier Data** Owns AigisQuery, the **React & Node.js** plain-English query platform analysts use themselves, on **Model Context Protocol (MCP)** tool servers under least-privilege access, reaching only cleared data.
 - Built the Agentic Harness Behind Production Workloads** Built the production-grade agent harness in **Node.js** and **Python** – loop detection, step budgets, prompt-injection defense, and a write allowlist – that every production agent runs on.
 - Cut Inference Cost by Leaving Frontier APIs** Moved production workloads onto in-house open-weight models (**DeepSeek, Qwen** on multi-GPU **vLLM**) with deterministic RAG grounding, cutting inference cost and driving SQL errors to **near zero**.
 - Held Queries Under 200ms Across Billions of Records** Scaled Node.js microservices with **Kafka, ClickHouse**, and **ElasticSearch** over billions of real-time spatial records, holding query latency under **200ms** at full production load.
- Apple** | *Software Development Engineer in Test, Core OS* Cupertino, CA | Jan 2023 – Jun 2024
 - Tripled Testing Throughput** Manual QA was the bottleneck on every **macOS** storage release. Migrated and rewrote 20+ aging test scripts in **Python & Node.js**, **tripling** testing run per cycle.
 - Protected 100M+ Macs from Storage Bugs** Worked **1,100+** high-priority tickets down to root cause so **APFS** filesystem updates went out stable across **100M+ Macs**.
 - Took the Manual Steps Out of Releases** Automated validation and release with **Ansible & Node.js**, cutting manual work **30%** and **tripling deployment speed**.
 - Stopped the Same Bugs Recurring** Turned recurring failure patterns into reusable, parameterized **Python** test suites, cutting duplicated test code **40%**.
 - Got Three Teams to Agree on Priorities** Product, engineering, and QA disagreed on priorities. Laid out what was broken and what it would cost, lifting throughput **20%**, and wrote troubleshooting guides that **halved** ramp time.
- Walmart** | *Software Engineer Intern* Sunnyvale, CA | May 2022 – Aug 2022
 - Tripled Backend Throughput** Rebuilt services in **Java, Spring Boot**, Node.js, and Hibernate on **Kubernetes** with **Kafka**, **tripling throughput** with Datadog and Grafana monitoring from day one.
 - Made Large Data Lookups Fast** Designed serverless **REST** and **gRPC** APIs on **AWS Lambda**, RDS, and **Redis** so large lookups returned faster and failed less often.
- Lawrence Berkeley National Laboratory** | *Software Engineer* Berkeley, CA | May 2021 – Sep 2021
 - Gave Researchers Back 200+ Hours a Year** Built a versioned pipeline in **Python, Docker**, and **Airflow** automating model tuning and recording every run, saving **200+ hours** annually.
- Honda Innovations** | *Software Engineer Intern* Mountain View, CA | Jan 2021 – May 2021
 - Automated Medical Delivery Routing** Built a model predicting fastest routes and shipped to **AWS** behind an automated **GitHub Actions, Docker**, and **Terraform** pipeline.

PROJECTS

- Open-Source Agent Tooling (76 public repos)** An AI agent is only worth as much as the systems it can safely reach. Publishes **6 Model Context Protocol servers** and **5 CLIs** that let agents drive mail, iMessage, WhatsApp, Notes, Jenkins, Bitbucket, Vercel, and Git worktrees, each scoped to only what it needs, shipped to real users through **Homebrew** and PyPI with a **Rust** hot path for local search.
- unbrowse – Web Data Without a Browser** Browser agents boot **Chromium** for every click, costing seconds and breaking on any DOM change. Wrote a tool that watches a site's own JSON calls once, writes a typed spec, and then talks straight to the network layer – **~40ms** per call instead of seconds – exposed as a CLI, a **REST** API, and an **MCP** tool.
- claude-tiers – Cheaper Long-Running Agents** Long autonomous runs get expensive because the frontier model writes the bulk code. Published a **Claude Code** plugin and ruleset that keeps planning and verdicts on the frontier model and routes implementation to cheaper ones, with evidence gates and an unattended overnight run mode, every rule graded against the literature.
- Multi-Agent Data and Research Pipelines** Built **CrewAI** pipelines where separate agents research, analyze, and audit each other's work to produce synthetic training data, backed by a Tokio pub/sub event bus in **Rust**.
- Document Agent That Resists Tampering** Invoices arrive as email attachments, and anyone can hide instructions inside one. Built an intake agent on the **OpenAI Agents SDK** that reads fields buried in embedded images while treating the document itself as untrusted.

PUBLICATIONS

6 peer-reviewed papers in machine learning and AI for environmental science ([Google Scholar](#))